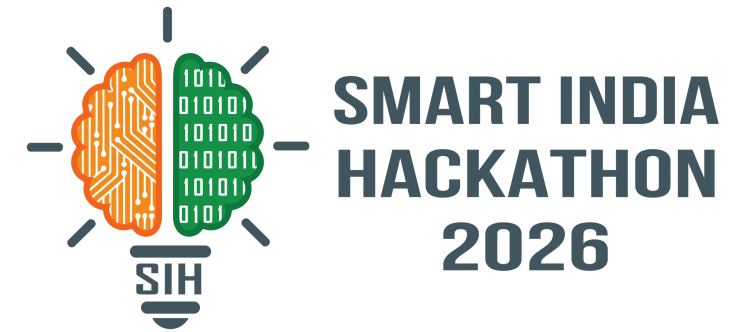


SMART INDIA HACKATHON 2026



Problem Statement ID: SIH26091

Problem Statement Title: AI-Driven Hyper-Local

Business Advisory and Financial Structuring

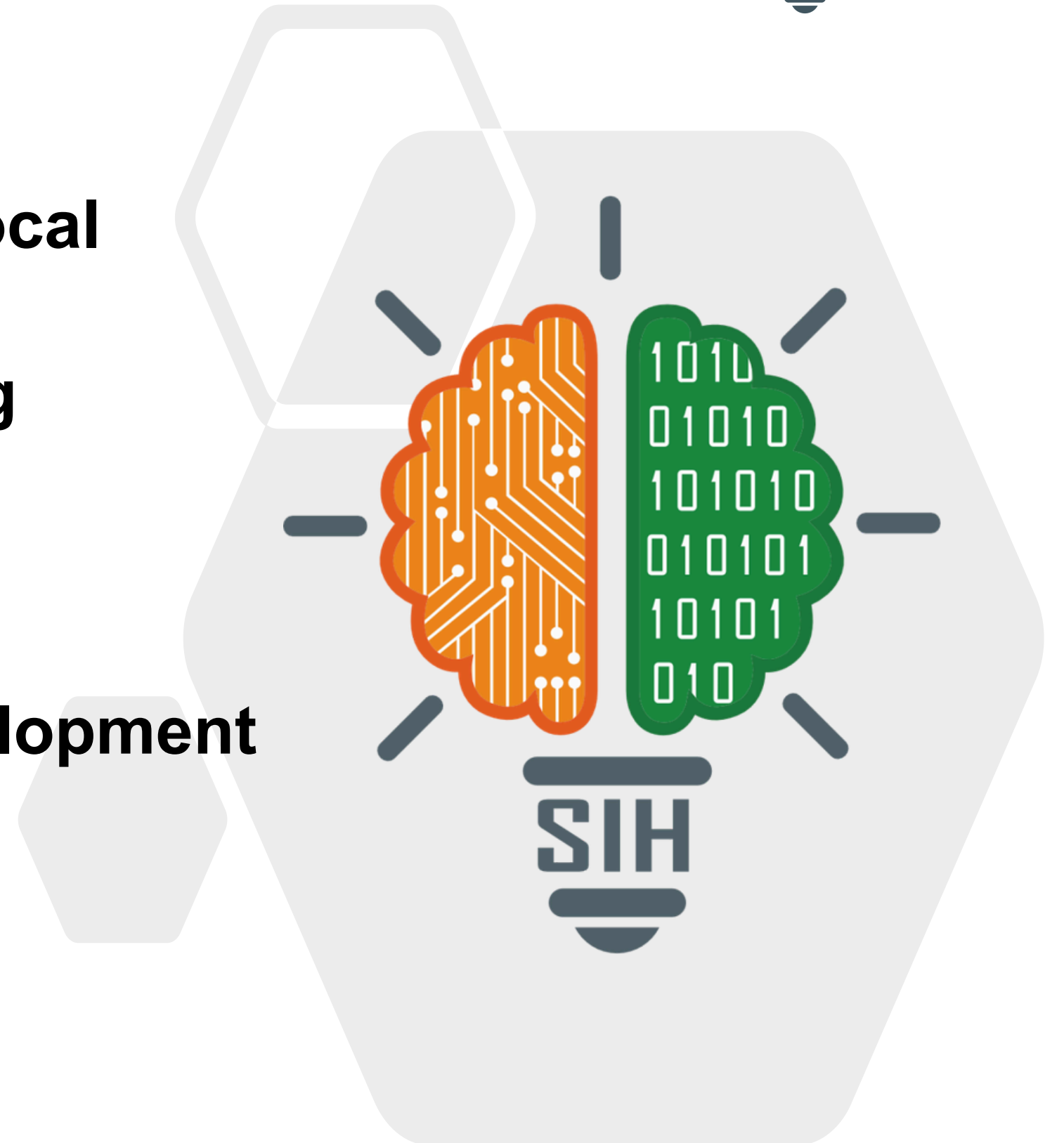
Assistant for Rural Micro-Entrepreneurs

Theme: Agriculture, FoodTech & Rural Development

PS Category: Software

Team ID: 137535

Team Name: Recursive Rebels



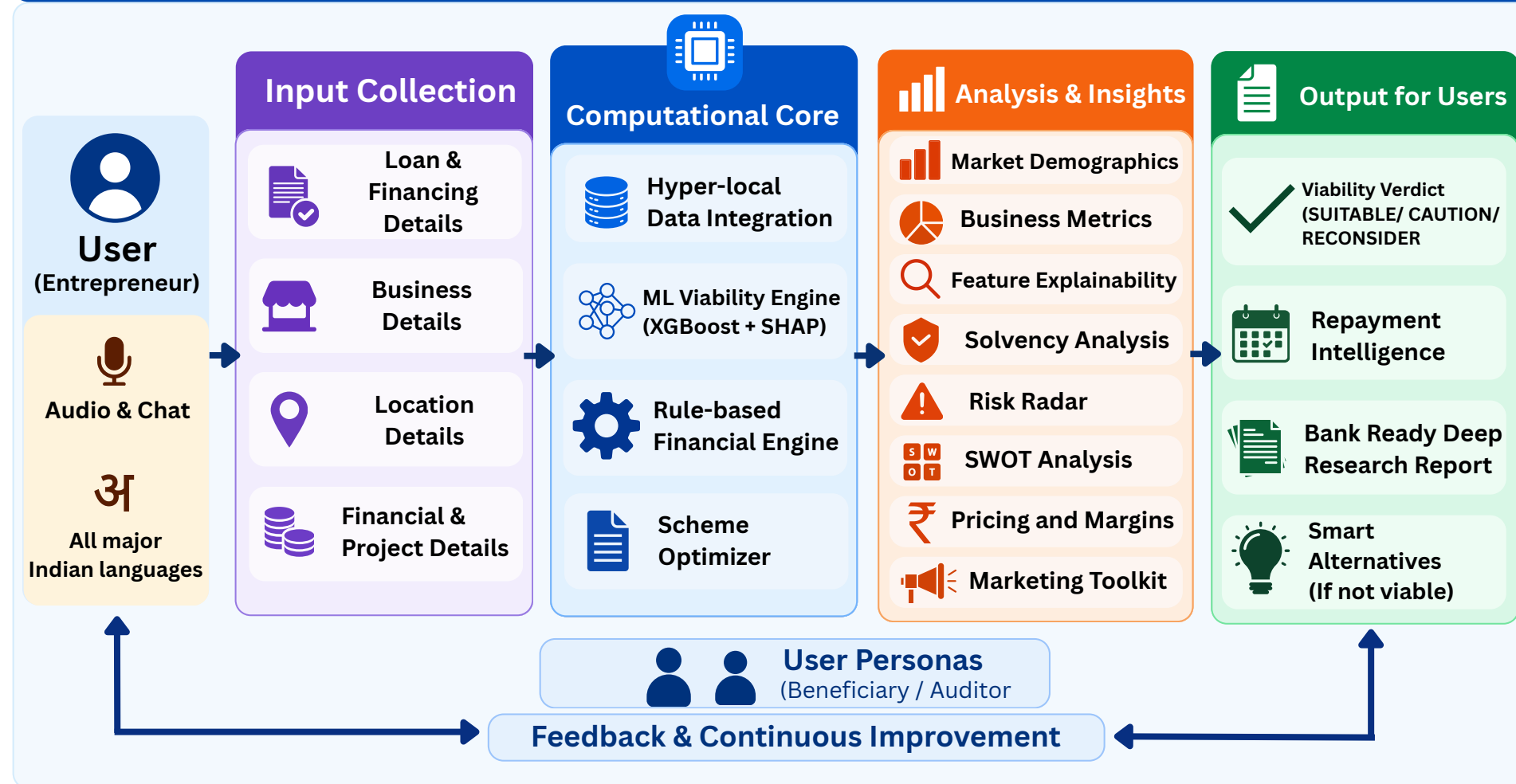
Udyam Saathi (उद्यम साथी) - AI Enterprise Feasibility & Statutory Bank DPR Engine

 **6.6L+ Villages • 788 Districts • 760+ ODOP**
Census CAGR • MSME Density • 613 Amenities • CPI **0% Hallucinated numbers • 22 Languages**
Bhashini Voice Control • Deterministic Banking Math **UN SDGs**
Ailigned     

Explanation of Proposed Solution

-  **Udyam Saathi Platform**
Feasibility, credit appraisal & bank DPR engine for rural/semi-urban MSMEs.
-  **Guided 7-Step Wizard**
Captures capex, location, equity & machinery with structured validation.
-  **Hyper-Local ODOP Core**
Census CAGR, village amenities & ODOP data to validate local market viability.
-  **Deterministic Finance Engine**
Nayak working capital, EMI schedules & RBI-mandated DSCR (≥ 1.33).
-  **17-Scheme Subsidy Optimizer**
Ranks PMEGP, PMFME, MUDRA to maximize capital grants (up to 35%).
-  **Bhashini Voice Assistant**
22 languages for verbal financial Q&A and voice-driven language switching.

User Flow Diagram



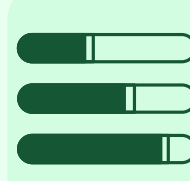
How it Addresses the Problem

-  **Slashes Bank Loan Rejections**
Replaces informal notes with statutory 7-Section DPRs meeting RBI norms.
-  **Resolves Market Blindspots**
Validates local demand gaps and competitor saturation before capital outlay.
-  **Unlocks Unclaimed Subsidies**
Matches 15%–35% capital grants across 17 schemes, bypassing middlemen.
-  **Clear risk assessment**
with **SUITABLE** / **CAUTION** / **RECONSIDER** verdicts
-  **Bridges Language Barrier**
Full vernacular speech guidance & voice navigation in 22 languages.

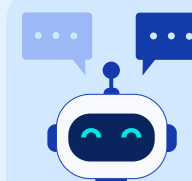
Innovation and Uniqueness of the Solution



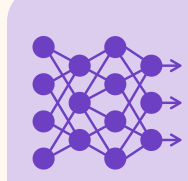
Counterfactual Viability Pivot
Detects village infrastructure strengths to recommend viable, bankable alternate business trades if initial idea is risky.



Lundberg TreeSHAP
Replaces black-box AI with exact game-theoretic factor attributions driving the viability verdict.



MIRA Voice Co-Pilot
Conversational agent answering financial queries and executing UI commands across 22 languages.



Zero-Hallucination Core
Deterministic algorithms compute 100% of banking math; the LLM only translates regional narrative summaries.

Recursive Rebels

TECHNICAL APPROACH

Data-Driven | Explainable | Accessible Technology



Users & Access

Anytime. Anywhere.



Rural Entrepreneur

- Business Ideas or Queries
- Local Context of the pinpoint location



Bank Loan Officer

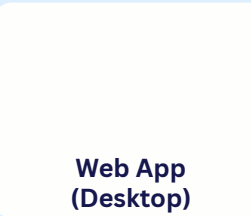
- Credit Appraisal
- Verification or Scheme Alignment



Govt. Nodal Agency

- Policy Monitoring
- Insights and Outreach Support

Multiple Access Channels



Web App (Desktop)



Mobile App (On-the-Go)



Voice Assistant (22 languages)



Multilingual Dashboard

Codebase, Demo & Prototype



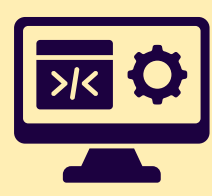
Codebase

CLICK HERE



Video

CLICK HERE



Prototype

CLICK HERE

```
graph TD
    subgraph Layer1 [1 Input Layer]
        direction LR
        L1_1[User Authentication]
        L1_2[User Query Business idea, location, budget]
        L1_3[Location details & context]
        L1_4[Tech Stack: React, VITE, Firebase]
    end

    subgraph Layer2 [2 Data Layer]
        direction LR
        L2_1[Government schemes & policies]
        L2_2[Hyperlocal & Regional data]
        L2_3[Indian Data Catalog]
        L2_4[Store and manage data]
        L2_5[Tech Stack: PostgreSQL, NEON]
    end

    subgraph Layer3 [3 Intelligence Layer]
        direction LR
        L3_1[Financial Engine]
        L3_2[Scheme Optimizer]
        L3_3[ML Models XGBoost Classifier]
        L3_4[Explainability Lundberg TreeSHAP]
        L3_5[LLM for explanation only]
        L3_6[Tech Stack: Shop, learn, BHASHINI, Anuvadini]
    end

    subgraph Layer4 [4 Decision Layer]
        direction LR
        L4_1[Policy Recommendations]
        L4_2[Cash Flow Projection]
        L4_3[Break Even Capacity]
        L4_4[DSCR Capacity]
        L4_5[Tech Stack: FastAPI, python]
    end

    subgraph Layer5 [5 Output Layer]
        direction LR
        L5_1[Detailed Project Report]
        L5_2[Financial Analysis]
        L5_3[Scheme Guidance]
        L5_4[Visual Insights & Dashboard]
        L5_5[Tech Stack: Render, docker, Vercel]
    end

    Layer1 -- "Validated and Structured Data" --> Layer2
    Layer2 -- "Integrated Data for Analysis" --> Layer3
    Layer3 -- "Model Outputs and Insights" --> Layer4
    Layer4 -- "Finalized Recommendations" --> Layer5
```

The diagram illustrates a five-layer technical approach for a rural entrepreneurship support system. It starts with an **Input Layer** for secure data capture, followed by a **Data Layer** for integration with government and regional data. The **Intelligence Layer** uses ML models and LLMs for analysis, leading to the **Decision Layer** which generates tailored insights like cash flow projections and capacity assessments. Finally, the **Output Layer** provides actionable recommendations through reports and dashboards. Each layer includes specific components and a dedicated tech stack.

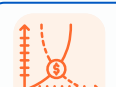
Outputs and Impacts



Detailed DPR (IBM Complaints)



Cash Flow Projections



Break Even Capacity



DSCR Capacity



Policy Recommendations

Accessible on All Devices

22 Indian Language Voice Assistant



22 Indian Language Voice Assistant

অসমীয়া (Assamese)

বাংলা (Bengali)

हिंदी (Hindi)

ગુજરાતી (Gujarati)

ଓଡ଼ିଆ (Oriya)

ਪੰਜਾਬੀ (Punjabi)

தமிழ் (Tamil)

తెలుగు (Telugu)

اردو (Urdu)

सिंधी (Sindhi)

बड़ो (Bodo)

ಕನ್ನಡ (Kannada)

മലയാളം (Malayalam)

मराठी (Marathi)

संस्कृतम् (Sanskrit)

नेपाली (Nepali)

कश्मीरी (Kashmiri)

मणिपुरी (Manipuri)

मेथिली (Maithili)

कोंकणी (Konkani)

संथाली (Santali)

डोगरी (Dogri)

Voice Agent Architecture



User Input (Voice)



Sarvam + Bhashini (Language Detection + ASR)



Speech-to-text (Transcription)



SARVAM 105B (Reasoning & Narration)



Response (Text)



Text-to-Speech (Bhashini)



Audio Output (User's Language)



Technical Feasibility

Proven + Explainable Architecture

- 98.9% XGBoost accuracy with TreeSHAP
- Deterministic Python math for RBI DSCR
- Zero-Recalculation invariant eliminates financial hallucinations
- Bank-grade reliability with end-to-end verified services



Economic Feasibility

Low-Cost Deployment

- < ₹0.50 DPR cost vs ₹15,000-₹50,000 CA fees
- 100% open data with zero licensing fee
- Sub-cent SARVAM inference + SHA-256 cache
- Usage-based cloud scaling



User Viability

Rural-First UX

- MIRA hands-free voice co-pilot in 22 languages
- Guided 7-step wizard with zero financial jargon
- Low-literacy UI with visual gauges & audio summaries
- Bank-Ready DPRs in Regional Languages



Operational Feasibility

Data Operations + Maintainance

- LGD 6-tier mesh auto-resolves village-to-state hierarchy
- Automated ingestion of 613 district amenities (Data.gov.in)
- Zero-crash offline fallbacks
- Dual-mode interface with specialized Credit Auditor inspection



Financial Viability

Data-Driven Regional Expansion

- B2B API monetization for pre-underwritten bank leads
- 95%+ gross margins with < ₹0.50 unit compute cost
- 100% free borrower access sponsored via Lead Bank / CSR
- Self-Sustaining at Zero Fixed Cost



Scalability

Ecosystem Adoption

- National DPI ready for Jansamarth & Udyam portals
- Stateless FastAPI containers for horizontal auto-scaling
- Serverless Neon PostgreSQL with async pooling
- Multi-Tenant Row-Level Security

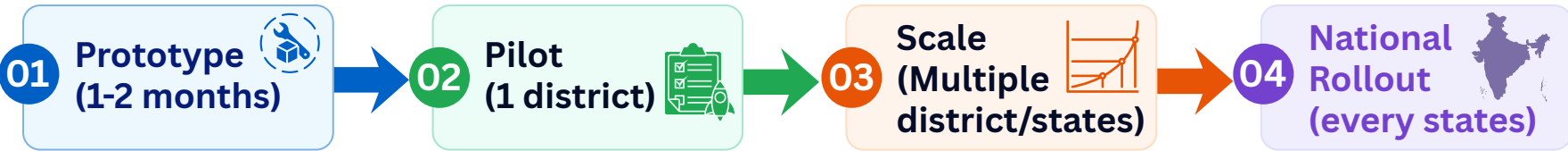


Feasibility Readiness Matrix

Dimension	Readiness Level	Key Evidence / Notes
Technical		Proven ML + API Stack
Data Availability		Open Government datasets + APIs
Deployment		Cloud Ready and modular system
Operational		Automated pipeline + monitoring
Economic		Usage based scaling is low cost
Scalability		Stateless service, easy to expand



Deployment and Scaling Path



Potential Risk and Challenges



Data Accuracy

Census 2011 Data Freshness
Outdated demographics may distort village demand estimates



Language & Extraction Variability

Rural Noise & Dialects
Noise, accents, and low bandwidth may affect voice recognition.



Scalability

GenAI Hallucinations
Financial inaccuracies risk banking liabilities.



Strategies for Overcoming

State CAGR projections to 2026 + MoSPI CPI + district MSME density.

BHASHINI noise-robust ASR + 7-step visual wizard fallback.

Zero-Recalculation Invariant:
Deterministic Python handles 100% of banking math.



Impact on Key Stakeholders

Empowering every stakeholder in the rural enterprise ecosystem



Rural & Micro-Borrowers

- Eliminates ₹25,000 - ₹50,000 DPR costs and unarredited middlemen dependence
- Provides professional business advice in 22 Indian languages via voice
- Maximizes subsidy access, enabling upto 35% applicable capital-linked support



Bank Loan Officers

- Cuts credit screening effort with pre-validated DPRs and financial metrics
- Makes risk transparent through SHAP-based explanations
- Standardized 7-Section IBA format cuts branch clerical workload by 80%.



Govt Nodal Agencies

- Boosts PMEGP, MUDRA, PMFME utilization through automatic eligibility matching
- Improves decision accuracy using grounded village-level data
- Speeds up sanctions by reducing processing bottlenecks and clerical workload



Multi-Dimensional Benefits

Inclusive Growth for a Stronger, Greener Rural Bharat



Economic Benefits

- ~15–25% reduction in transaction costs through digitized business appraisal and financing workflows.
- Prevents undercapitalization & debt distress through automated capital reconciliation
- Reduces Banking NPAs via RBA benchmark-based debt solvency screening
- 10–20% faster capital deployment through pre-validated DPRs and streamlined credit processing.



Commercial Benefits

- ~30–40% lower CAC through a digital entrepreneur-to-lender acquisition funnel.
- 5-year projections expose cash-flow gaps before they become funding problems.
- 760+ ODOP categories enable data-backed diversification into local opportunities.
- 26% leaner data footprint for downstream processing.



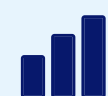
Social Benefits

- Democratizes financial literacy across 22 Indian languages
- Enforces statutory 5% equity concessions for women, SC/ST & Artisan founders
- Removes literacy and typing barriers through multilingual voice AI Agent
- Creates sustainable, dignified non-farm livelihoods inside villages (SDG 11).



Environment Benefits

- Operates 100% paperless, across the appraisal and DPR lifecycle
- Eliminates multiple >20 km bus/bike trips from villages to district bank branches.
- Replaces physical couriering and multi-copy printing with digital delivery
- Uses village infrastructure readiness to support sustainable rural enterprises

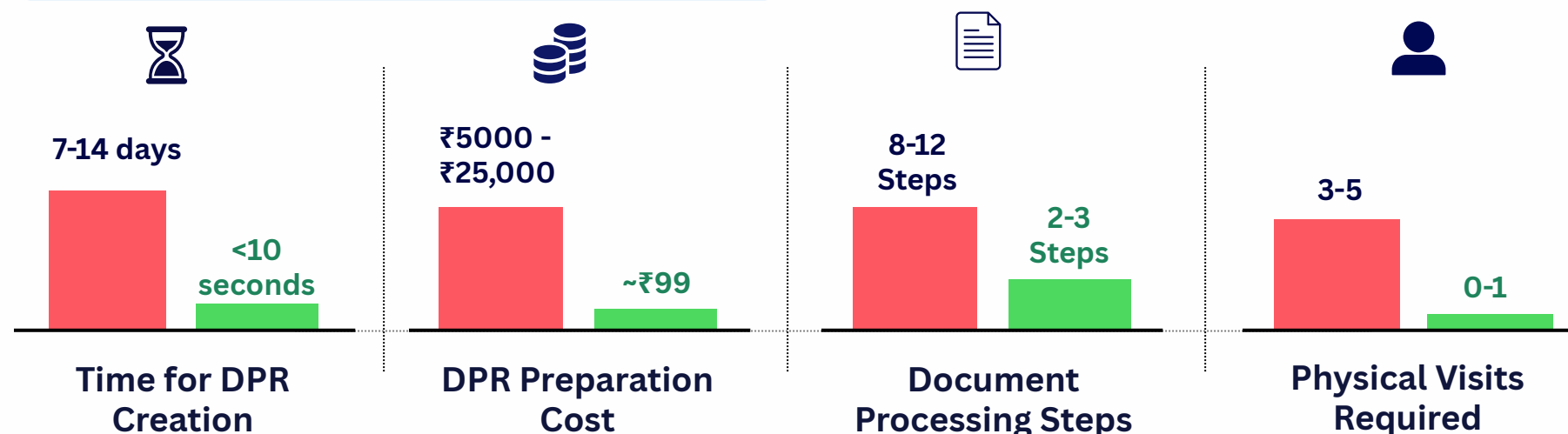


Process Efficiency Comparison

From weeks of paperwork to seconds of possibilities

Current Process

Our Solution

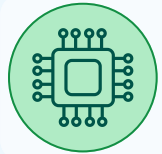


RESEARCH AND REFERENCES

Trusted Knowledge | Real-world Data | Stronger Solution



A flagship initiative by the Government of India to democratize AI for inclusive growth, public good and societal empowerment



Compute Access

High Performance GPU infrastructure (AIRAWAT)



AI Innovation Ecosystem

Dataset, models and tools for indian languages & use



Skill & Talent Development

Building an AI-ready workforce for Bharat



Application for Social Good

AI solutions for agriculture, health, education, governance and rural development



Guides our approach towards responsible, inclusive and impact driven AI solutions

indiaai.gov.in

